

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-35

Faculty

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Operational Amplifiers (op-amp)

- An op amp is an active circuit element which can sum signals, amplify a signal, integrate it, or differentiate it. The ability of the circuit to perform these mathematical operations is the reason it is called an operational amplifier.
- An op amp may also be regarded as a voltage amplifier with very high gain.

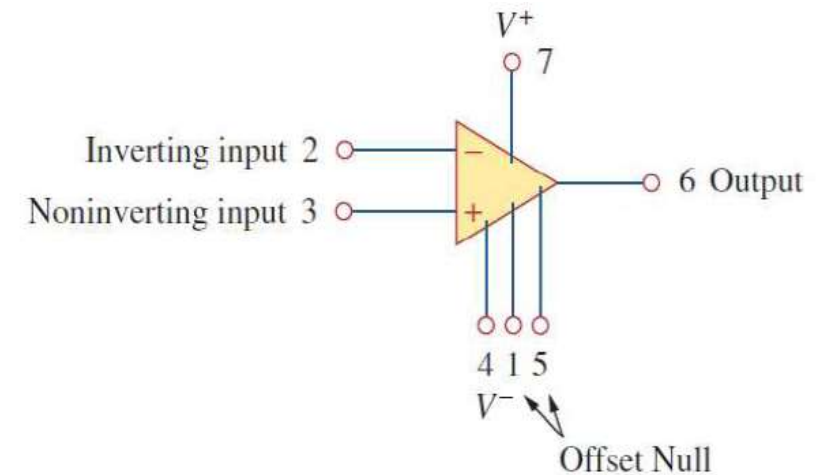
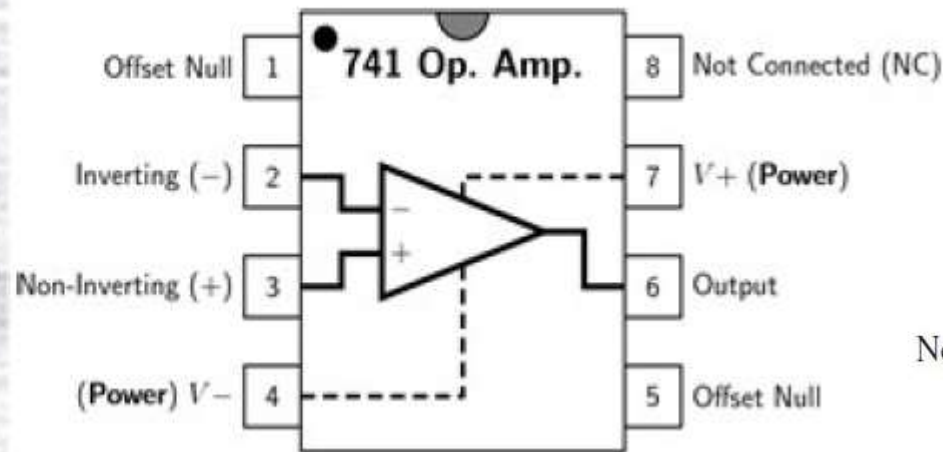
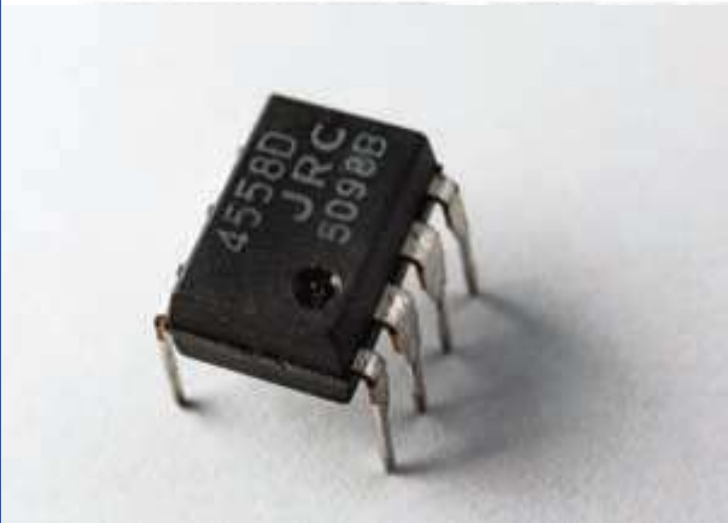
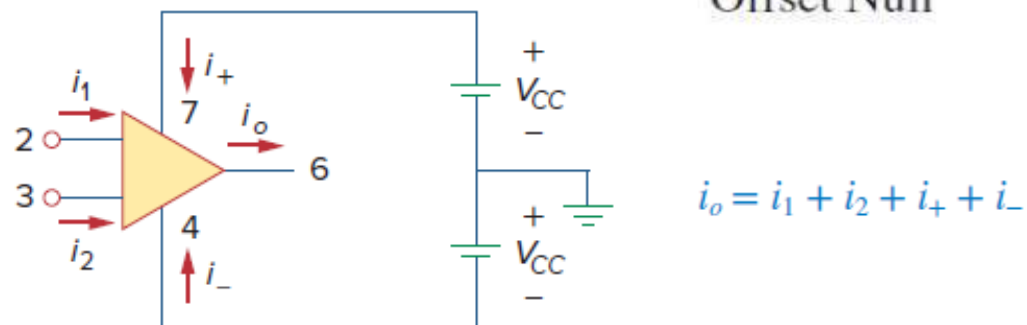
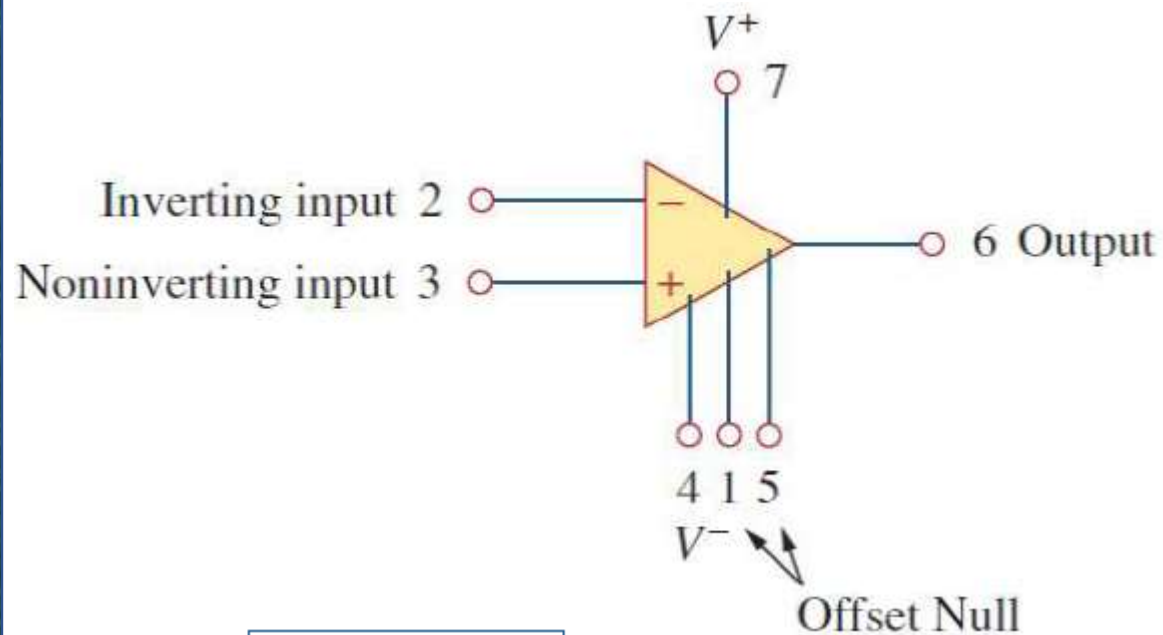
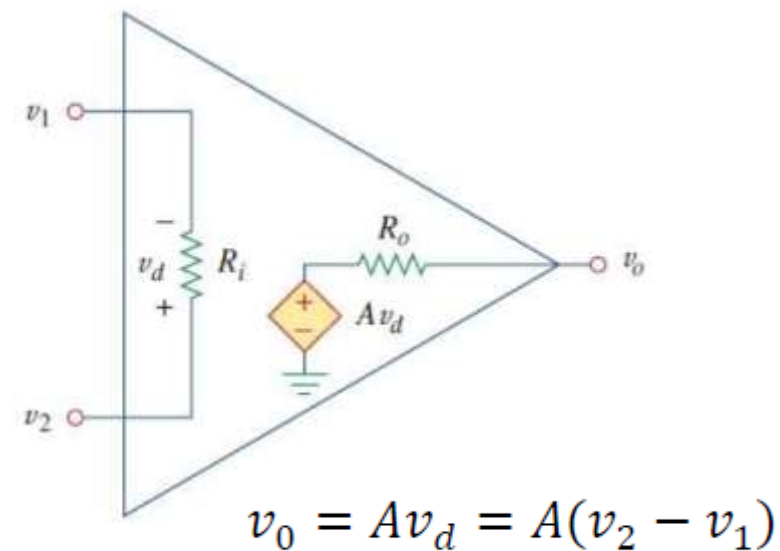


Figure: A typical operational amplifier and pin diagram

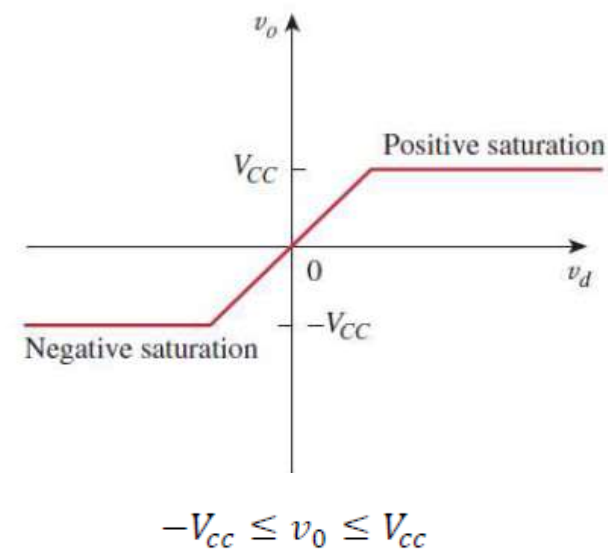


$$i_o = i_1 + i_2 + i_+ + i_-$$

Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to 10^8	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100Ω	0Ω
Supply voltage, V_{CC}	5 to 24 V	



$$v_o = A v_d = A(v_2 - v_1)$$



$$-V_{CC} \leq v_o \leq V_{CC}$$

1) Inverting Amplifier

- An amplifier which produces a phase shift of 180° between input and output is called inverting amplifier. Fig. shows an inverting amplifier using op-amp.

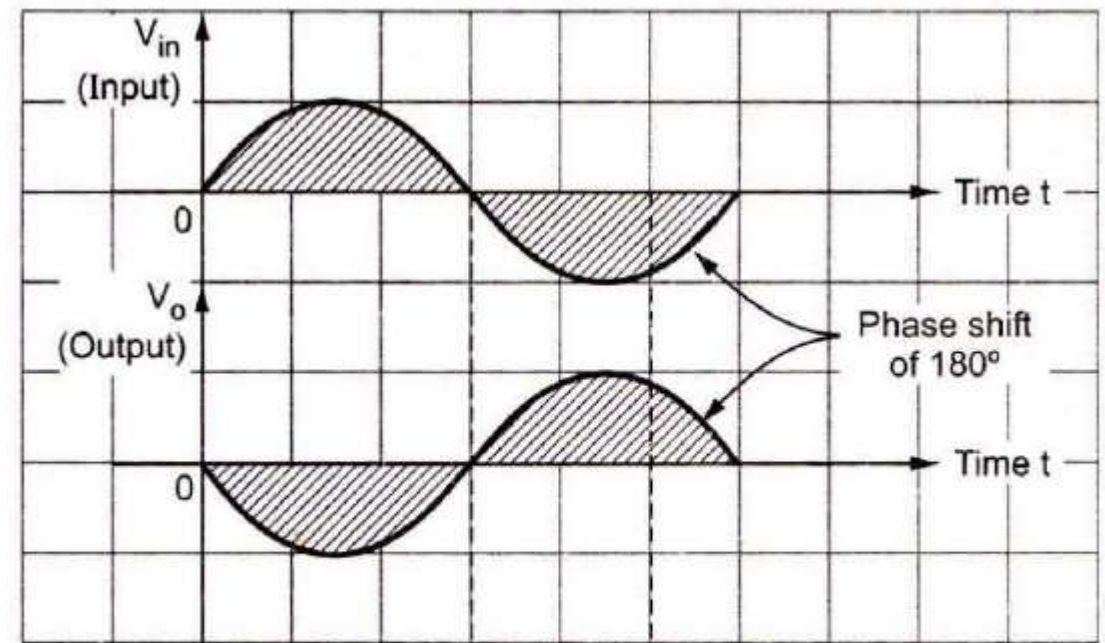
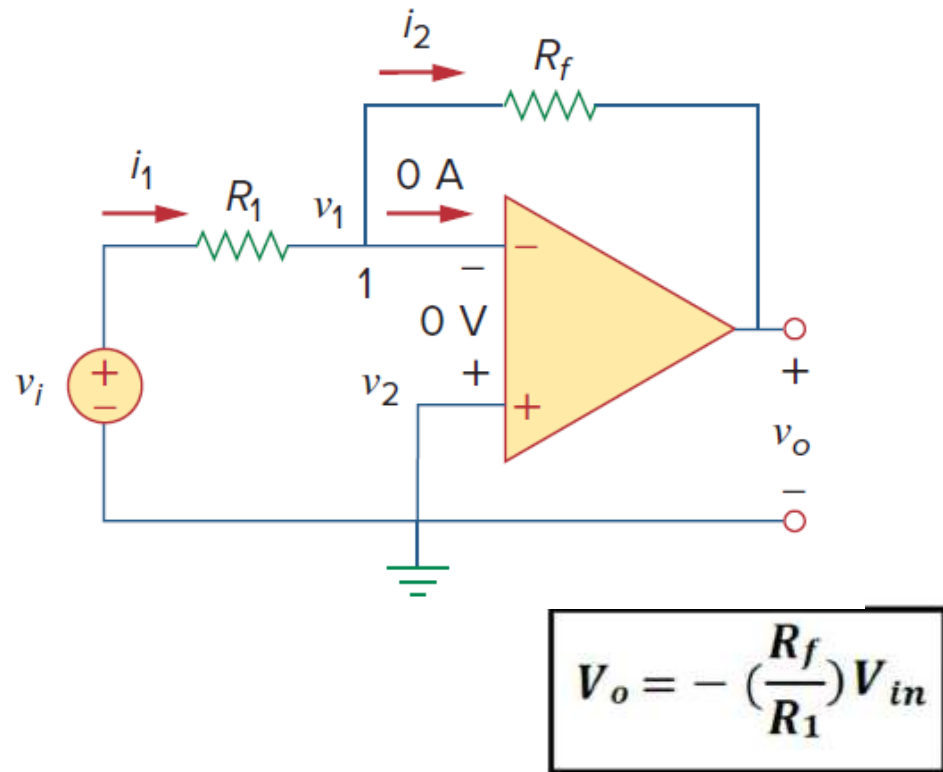
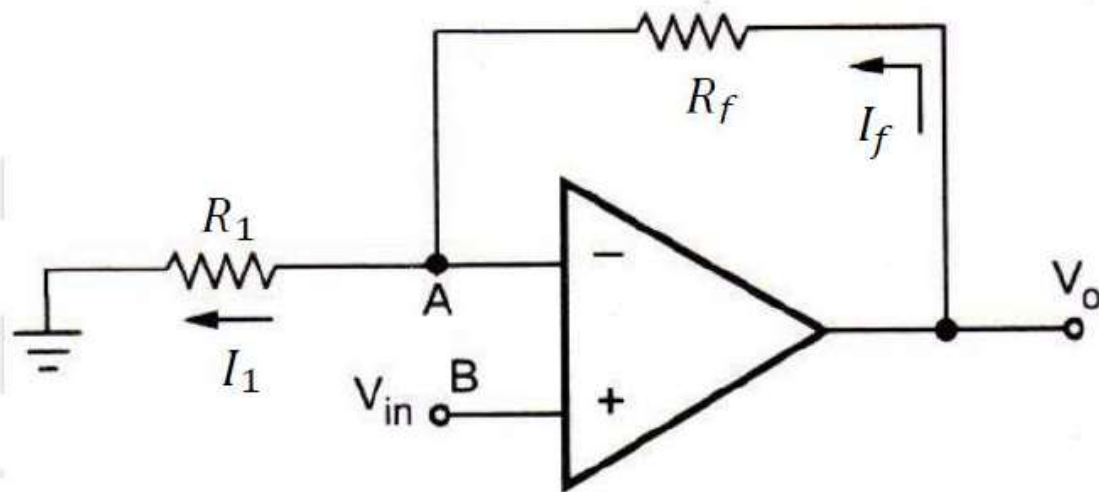


Fig. Waveforms of inverting amplifier

2) Non-Inverting Amplifier

- An amplifier which amplifies the input without producing any phase shift between input and output is called non-inverting amplifier.



$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

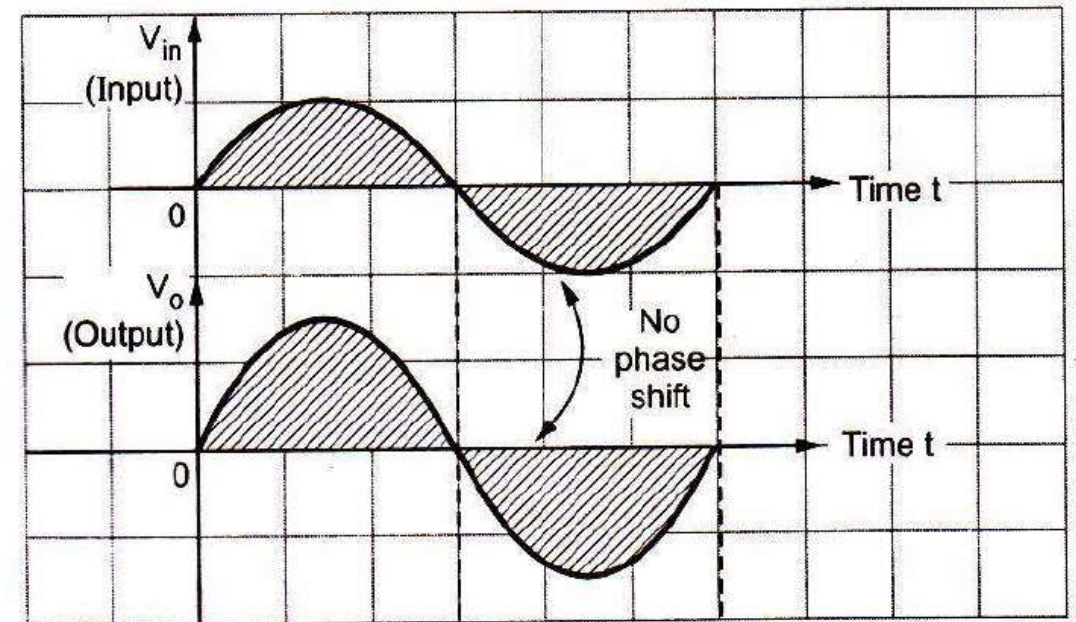
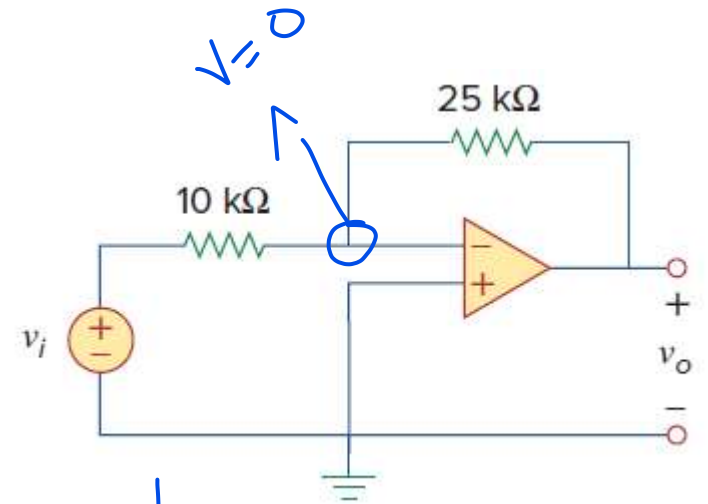


Fig. Waveforms of non-inverting amplifier

1) Refer to the op amp in Fig. below. If $v_i = 0.5$ V, calculate: (a) the output voltage v_o , and (b) the current in the 10-k Ω resistor.



Inverting

$$A = -\frac{R_f}{R_i} = -\frac{25}{10} = -2.5$$

$$V_o = 2.5 \times 0.5$$

$$V_o = 1.25 \text{ V}$$

$$I_{10} = \frac{0.5}{10}$$

$$I_{10} = 0.05 \text{ mA}$$